* Revision for Week 3

(Ω, F, P) — sample space, collection of events, probability measure

* Random Variable : $X : \Omega \longrightarrow \mathbb{R}$
* Discrete Random Variable — S_X is finite or countable
* Cumulative Distribution Function (CDF) $F_X(x) = P(X \leq x)$
* Probability mass function (PMF) $P_X(i) = P(X=i)$ $\xrightarrow{x \in \mathbb{R}}$ $i \in S_X \subseteq \mathbb{R}$
  - Bernoulli Random variable → $\{0,1\}$ $P(X=1)=p$ $P(X=0)=1-p$
  - Binomial Random Variable → $\{0,1,2,\ldots,n\}$ $P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$
  - Geometric Random Variable → $P(X=k) = (1-p)^{k-1} p$
  - Poisson Random Variable → $\boxed{P(X=k) = e^{-\lambda} \frac{\lambda^k}{k!}}$, $\lambda > 0$
* Examples of Binomial R.V.
* Continuous Random Variable
  - Pdf $f_X : \mathbb{R} \to [0, \infty)$
* Uniform Random variable → $f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{o.w.} \end{cases}$
* Exponential Random Variable → memoryless property.
* Gaussian Random Variable

Uniform random variable $U[0,1]$ (a, b)

$$f_X(x) = \begin{cases} 1 & \text{if } 0 \leq x \leq 1 \\ 0 & \text{o.w.} \end{cases}$$

$\boxed{P[X=x] = 0}$

$$P\left[\tfrac{1}{2} \leq x \leq \tfrac{3}{4}\right] = \int_{1/2}^{3/4} f_X(x)\, dx = \int_{1/2}^{3/4} 1\, dx = [x]_{1/2}^{3/4} = \tfrac{3}{4} - \tfrac{1}{2} = \tfrac{1}{4}$$

$$P\left[-\tfrac{1}{2} \leq x \leq \tfrac{1}{2}\right] = \int_{-1/2}^{1/2} f_X(x)\, dx = \int_{-1/2}^{0} f_X(x)\, dx + \int_{0}^{1/2} f_X(x)\, dx$$

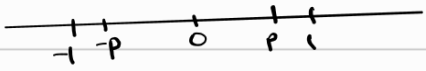
$$= \int_{-1/2}^{0} 0\, dx + \int_{0}^{1/2} 1\, dx$$

$$= 0 + \tfrac{1}{2} = 1/2$$

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Problem Set

16. Consider two events A and B such that $\mathbb{P}(A) > 1 - \delta$, and $\mathbb{P}(B) > 1 - \delta$, for very small $\delta > 0$. Prove that $\mathbb{P}(A \cap B)$ is close to 1.



$p \leq 1$

$-p \geq -1$

$0 \leq P(A \cup B) \leq 1$

$-P(A \cup B) \geq -1$

$P(A) > 1-\delta$

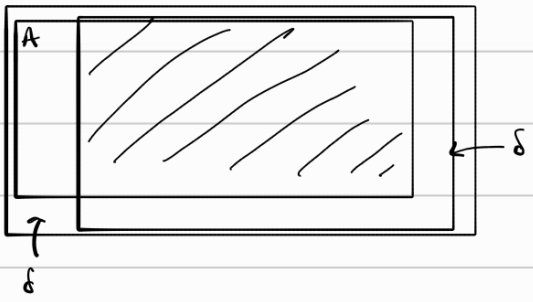
$P(B) > 1-\delta$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cap B) = P(A) + P(B) + (-P(A \cup B))$

$P(A \cap B) > 1-\delta + 1-\delta + (-1)$

$P(A \cap B) > 1 - 2\delta$



2δ is also small

$\therefore P(A \cap B) \approx 1$

$P(A \cap B) = P(A) + P(B) + (-P(A \cup B)) \geq (1-\delta) + (1-\delta) + (-1)$

$P(A \cap B) > 1 - 2\delta$

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2. Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times. What are the possible values of X?
3. In Problem 2, if the coin is assumed fair, for $n = 3$, what are the probabilities associated with the values that X can take on?

$X : \Omega \to \mathbb{R}$

Guess 1: $-n$ to n $\quad (n-1)$

$+2$

k heads

$n-k$ tails

$\to k - (n-k)$

$\to -n + 2k$

n is even

$S_X = \{0, \pm 2, \pm 4, \ldots, \pm n\}$

n is odd

$S_X \{\pm 1, \pm 3, \ldots, \pm n\}$

$n = 3$ $\quad S_X = \{\pm 1, \pm 3\}$

$$P_X(i) = \begin{cases} 1/8 & i = -3 \\ 3/8 & i = -1 \\ 3/8 & i = 1 \\ 1/8 & i = 3 \end{cases}$$

HTT

THT

TTH

$n = 4$

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EXAMPLE 2a

The probability mass function of a random variable X is given by $p(i) = c\lambda^i/i!$, $i = 0, 1, 2, \ldots$, where λ is some positive value. Find (a) $P\{X = 0\}$ and (b) $P\{X > 2\}$.

$$p(i) = c\,\frac{\lambda^i}{i!}$$

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$\Rightarrow \sum_{i=0}^{\infty} c\,\frac{\lambda^i}{i!} = 1$$

$$\Rightarrow c\left(\sum_{i=0}^{\infty} \frac{\lambda^i}{i!}\right) = 1$$

$$c \cdot e^{\lambda} = 1$$

$$\therefore c = e^{-\lambda}$$

$$\boxed{p(i) = e^{-\lambda}\frac{\lambda^i}{i!}} \quad \text{for} \quad i = 0, 1, 2, \ldots$$

$$P(X=0) = e^{-\lambda}$$

$$P(X>2) = 1 - P(X \le 2) = 1 - (P(X=0) + P(X=1) + P(X=2))$$

$$= 1 - e^{-\lambda} - e^{-\lambda}\lambda - e^{-\lambda}\frac{\lambda^2}{2}$$

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5. At least one-half of an airplane's engines are required to function in order for it to operate. If each engine independently functions with probability p, for what values of p is a 4-engine plane more likely to operate than a 2-engine plane?

X as number of working engines in 4-engine plane

$0, 1, 2, 3, 4$

$X \sim Bin(4, p)$ atleast 2 engines functioning

2-engine plane $Y \sim Bin(2, p)$ atleast 1 engine functioning

$$\boxed{P(X \geq 2) > P(Y \geq 1)}$$

$(1-p)^4$ $\quad \binom{4}{1} p^1 (1-p)^3$

$$P(X \geq 2) = 1 - P(X < 2) = 1 - P(X=0) - P(X=1)$$
$$= 1 - (1-p)^4 - 4p(1-p)^3 \rightarrow$$

$$P(Y \geq 1) = 1 - P(Y < 1) = 1 - P(Y=0)$$
$$= 1 - (1-p)^2 \rightarrow$$

$$1 - (1-p)^4 - 4p(1-p)^3 > 1 - (1-p)^2$$
$$(1-p)^4 + 4p(1-p)^3 < (1-p)^2$$
$$(1-p)^3 (1-p+4p) < (1-p)^2$$
$$(1-p)(1+3p) < 1$$
$$1 - p + 3p - 3p^2 < 1$$
$$2p - 3p^2 < 0$$
$$p(\underbrace{2-3p}_{<0}) < 0 \rightarrow$$

$2 - 3p < 0 \rightarrow 2 < 3p \rightarrow \frac{2}{3} < p$

$$\boxed{p > \frac{2}{3}}$$

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